29/08/2026

Code-C Phase-3

Aakash

Medical|IIT-JEE|Foundations

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MM : 720

Meridian Test for NEET-2027 (XII-Passed_P3)_Test-01C

Time : 180 Min.

Topics Covered:
Physics: Units & Measurements, Motion in a Straight Line, Motion in a Plane, Laws of Motion
Chemistry: Some Basic Concepts of Chemistry, Structure of Atom, Classification of Elements and Periodicity in Properties
Botany: Cell: The Unit of Life, Cell Cycle & Cell Division, The Living World, Biological Classification-I: (Upto thermoacidophiles)
Zoology: Structural Organisation in Animals– Animal Tissues, Biomolecules, Breathing & Exchange of Gases-I: (Upto mechanism of breathing)

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. The velocity of a particle varies with distance x from a fixed origin as $v = \frac{A\sqrt{x}}{x+B}$ where A and B are constants. The dimensional formula of AB is
(1) $[M^0 L^{5/2} T^{-1}]$ (2) $[M^0 L^{3/2} T^{-2}]$
(3) $[M^0 L^{1/2} T^{-1}]$ (4) $[M^0 L^{3/2} T^{-1}]$

2. The displacement of a particle moving with constant acceleration is, $s = ut + at^2$ (where u is velocity in m/s, a is acceleration in m/s^2 and t is time in second). The equation is
(1) Dimensionally true (2) Numerically true
(3) Dimensionally false (4) Both (1) and (2)

3. In a vernier calliper 20 vernier scale divisions are equal to 16 main scale divisions. If one main scale division is equal to 2 mm, then what is least count of the instrument?
(1) 0.2 mm (2) 0.1 mm
(3) 0.4 mm (4) 0.5 mm

4. Four physical quantities are given. Which is the correct order of ascending dimension of time in them?
A) Linear momentum
B) Force
C) Power
D) Density
(1) D, A, C, B (2) C, B, A, D
(3) D, C, B, A (4) A, C, B, D

5. If velocity (v), time (T) and mass (M) are taken as fundamental quantities, then dimensional formula for gravitational potential would be
(1) $[MvT^0]$ (2) $[Mv^2T^0]$
(3) $[M^0v^2T^0]$ (4) $[M^{-2}vT^0]$

6. In a certain system, the units of length, mass and time are chosen to be 1 cm, 100 g and 10 s. The unit of kinetic energy will be
(1) 0.1 erg (2) 10 erg
(3) 1 erg (4) 20 erg

7. The radius of a sphere is (20.0 ± 0.2) cm. The percentage error in its surface area is
(1) 4% (2) 3%
(3) 2% (4) 1%

8. Which among the following relation cannot be deduced by dimensional method?
(1) $y = A\cos(\omega t - kx)$ (2) $S = ut + \frac{1}{2}at^2$
(3) $K = \frac{1}{2}mv^2$ (4) All of these

9. If random error in arithmetic mean of 200 observations is e, then random error in arithmetic mean of 600 observations would be
(1) $4e$ (2) e
(3) $\frac{e}{3}$ (4) $\frac{e}{4}$

10. Dimensions of stress are :
(1) $[ML^2T^{-2}]$ (2) $[ML^0T^{-2}]$
(3) $[ML^{-1}T^{-2}]$ (4) $[MLT^{-2}]$

11. A particle is moving with uniform acceleration. Its position (x) is given in terms of time t (in s) as $x = (5t^2 + 4t + 8)$ m, then
(1) Its initial velocity is 8 m/s
(2) Its velocity at time $t = 3$ s is 36 m/s
(3) Its acceleration is 5 m/s^2
(4) Its position at $t = 2$ s is 36 m

12. A body covers one-third of the distance with a speed v_1, the second one-third of the distance with a speed v_2 and the last one-third distance with a speed v_3. The average speed of the body for whole distance is

(1) $\frac{3v_1v_2v_3}{v_1v_2+v_2v_3+v_3v_1}$

(2) $\frac{v_1v_2+v_2v_3+v_3v_1}{v_1+v_2+v_3}$

(3) $\frac{v_1+v_2+v_3}{3}$

(4) $\sqrt{\frac{v_1^2+v_2^2+v_3^2}{2}}$

13. The initial velocity of a particle is u and the acceleration a is given by $a = 2t^{3/2}$. The velocity of the particle at any time t is

(1) $v = u + \frac{4}{5}t^{1/2}$ (2) $v = u + \frac{2}{5}t^{3/2}$

(3) $v = u + \frac{4}{5}t^{5/2}$ (4) $v = u + \frac{2}{5}t^{1/2}$

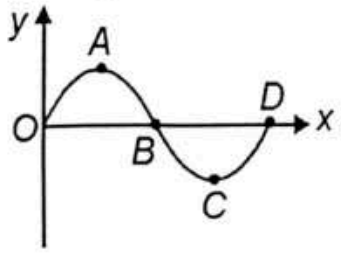
14. The linear momentum P of a body moving in one dimension varies with time according to equation $P = a + bt^2$, where a and b are constant. The force acting on the body is

(1) Constant
(2) Proportional to square of time
(3) Inversely proportional to time
(4) Directly proportional to time

15. The velocity of particle moving along positive x-axis varies with position x as $v = 2\sqrt{x}$, where v is in m/s and x is in m. The average velocity of particle for the motion from $x = 0$ to $x = 4$ m is

(1) 0.5 m/s (2) 0.25 m/s
(3) 1 m/s (4) 2 m/s

16. At which point (s) the value of $\frac{dy}{dx} = 0$ for the given graph as shown in figure?



(1) A (2) B
(3) C (4) Both (1) & (3)

17. If the velocity of a particle is given as $v = x^{3/2}$, then acceleration versus displacement graph is (Where x is displacement of particle)

(1) Parabola (2) Circle
(3) Straight line (4) Rectangular hyperbola

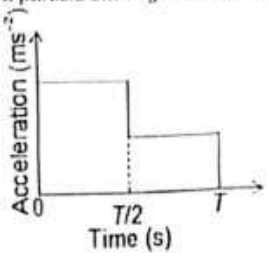
18. When a ball is thrown vertically upwards with initial velocity of 20 m/s, what will be the maximum height it will reach and when will it be? (Take $g = 10$ m/s^2)

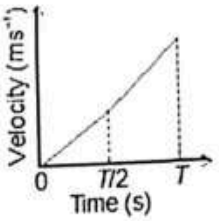
(1) 20 m, 2 s (2) 40 m, 4 s
(3) 45 m, 3 s (4) 25 m, 5 s

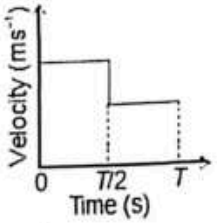
19. A ball is dropped from the top of a building. If it covers $\left(\frac{3}{4}\right)^{th}$ of its total height in the last second of its journey, then total time of fall is

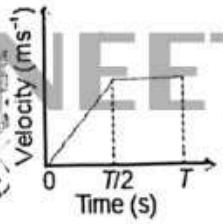
(1) 1.5 s (2) 2.5 s
(3) 2 s (4) 3 s

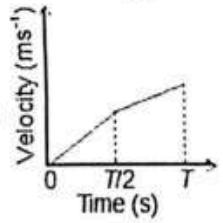
20. For the given acceleration-time graph, velocity-time graph of a particle starting from the rest is



(1)


(2)


(3)


(4)


21. A car travels first 10 seconds with uniform acceleration of 2 m/s^2 from rest on straight road and then travels with zero acceleration for next 10 seconds. The average speed of entire journey is

(1) 20 m/s (2) 15 m/s
(3) 10 m/s (4) 5 m/s

22. The equation of a projectile in x-y plane, projected at an angle θ w.r.t +ve x direction is given by $y = 80x - 5x^2$. The maximum height attained by the projectile is (Here x and y are in meter)

(1) 160 m (2) 320 m
(3) 240 m (4) 120 m

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23. A : A unit vector has unit magnitude.
R : A unit vector may not have direction associated with it.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

24. Two particles are revolving in circular paths of radii 4 cm and 12 cm with same constant speed. The ratio of magnitudes of their centripetal accelerations is
(1) 3 : 1 (2) 1 : 2
(3) $1:\sqrt{3}$ (4) $\sqrt{3}:1$

25. If a_r and a_t represents the magnitude of radial and tangential acceleration, the motion of the particle will be uniform circular motion if
(1) $a_r = 0, a_t = 0$ (2) $a_r = 0, a_t \neq 0$
(3) $a_r \neq 0, a_t \neq 0$ (4) $a_r \neq 0, a_t = 0$

26. Two projectiles *A* and *B* are projected with same speed at angles 15° and 75° respectively to the horizontal on ground. Which of the following is correct? (T - time period, H - maximum height)
(1) $H_A > H_B$ (2) $H_A < H_B$
(3) $T_A < T_B$ (4) Both (2) and (3)

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27. A train moves northwards with speed 80 km h^{-1}, while a car moves towards east with speed of 60 km h^{-1}. What is magnitude of velocity of train w.r.t. driver of the car?
(1) 140 km h^{-1} (2) 20 km h^{-1}
(3) 100 km h^{-1} (4) 70 km h^{-1}

28. A particle starts from origin and moves with a constant acceleration of $\left(7\hat{i}+3\hat{j}\right)m/s^2$.
The position vector of the particle at $t = 2$ s will be
(1) $\left(14\hat{i}-3\hat{j}\right)m$ (2) $\left(-14\hat{i}-3\hat{j}\right)m$
(3) $\left(7\hat{i}+6\hat{j}\right)m$ (4) $\left(14\hat{i}+6\hat{j}\right)m$

29. If two non-zero vectors $\vec{A}$ and $\vec{B}$ obey the law $|\vec{A}-\vec{B}| = |\vec{A}-\vec{B}|$, then the angle between $\vec{A}$ and $\vec{B}$ is
(1) 120° (2) 90°
(3) 60° (4) 0°

30. A particle projected from earth surface is moving in vertical plane (*x*-*y* plane) such that its trajectory is given by the equation, $y = x - \frac{x^2}{80}$, where *x* & *y* are in metre.
For this particle match column-I with column-II and tick the correct option.

	Column A		Column B
A.	Horizontal range (m)	(P)	45
B.	Angle of projection with horizontal (degree)	(Q)	80
C.	Maximum height gained by particle (m)	(R)	20
D.	Speed of projection (m s^{-1})	(S)	$20\sqrt{2}$

(1) A(S), B(R), C(Q), D(P) (2) A(P), B(Q), C(S), D(R)
(3) A(Q), B(P), C(S), D(R) (4) A(Q), B(P), C(R), D(S)

31. At a height of 0.6 m from the ground, the velocity of projectile in vector form is $\vec{v} = \left(6\hat{i}+2\hat{j}\right)m/s$ (the *x*-axis is horizontal and *y*-axis is vertically upwards). The initial speed of the projectile will be (take $g = 10$ m/s^2)
(1) 10 m/s (2) 14 m/s
(3) $\sqrt{42}$ m/s (4) $\sqrt{52}$ m/s

32. If a vector $\overrightarrow{OP} = 3\hat{i}+3\hat{j}$ is turned clockwise by an angle of 15^0, then the y-component of rotated vector would be
(1) 6 (2) $\frac{3}{\sqrt{2}}$
(3) $\frac{2}{3}$ (4) $\frac{3}{2}$

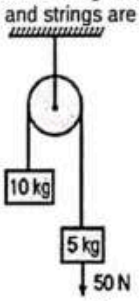
33. Read the following statements carefully and choose the correct option
Statement(A): A man can not walk by himself on a frictionless horizontal surface
Statement(B): In the absence of friction, we can not push the surface tangentially backward.
(1) Both statements (A) and (B) are correct
(2) Both statements (A) and (B) are incorrect
(3) Statement (A) is correct but statement (B) is incorrect
(4) Statement (A) is incorrect but statement (B) is correct

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34. An aircraft moving in a horizontal circular loop at a speed of 360 kmph. If its wings are banked at an angle 53°, then diameter of circular loop is (take $g = 10$ m/s^2)
(1) 1 km (2) 0.75 km
(3) 1.5 km (4) 2 km

35. Two men of masses *m* and *m*/2 start climbing up on two separate massless strings fixed at the ceiling with acceleration *g* and *g*/2 respectively. The ratio of tensions in the two corresponding strings will be
(1) 2 : 1 (2) 4 : 1
(3) 4 : 3 (4) 8 : 3

36. The acceleration of the blocks and tension in the string connecting both blocks, are respectively (assuming pulley and strings are ideal)



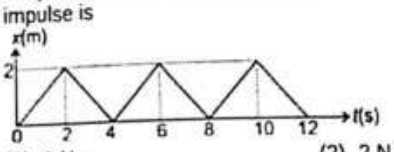
(1) 2 m/s^2, 100 N (2) 5 m/s^2, 100 N
(3) Zero, 100 N (4) Zero, 200 N

37. A block of mass *M* is pulled along a horizontal frictionless surface by a rope of mass *m*. If a horizontal force *P* is applied at the free end of the rope, the force exerted by the rope on the block will be
(1) *P* (2) $\frac{P_m}{M+m}$
(3) $\frac{PM}{M+m}$ (4) $\frac{Pm}{M+m}$

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38. **Statement A:** If net external force acting on a body is parallel or anti-parallel to its motion, then path followed by the body is a straight line.
Statement B: If net external force is always perpendicular to the motion of body, then path followed by the body is circular.
Statement C: If net external force acts at an angle to the motion of the body, then path followed by the body may be parabolic or non-uniform circular motion.
Consider the given statements and choose the correct option.
(1) Only statement A is true
(2) Only statement B is true
(3) Only statement C is true
(4) All statements are true

39. The position(x)-time(t) graph of one dimensional motion of a body of mass 0.5 kg is shown. The magnitude of each impulse is

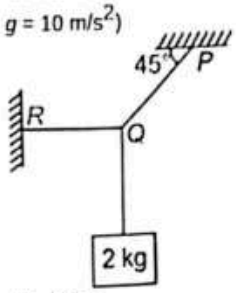


(1) 1 N s
(2) 2 N s
(3) 0.5 N s
(4) 1.5 N s

40. Which of the following statements is not correct about impulse?
(1) Impulse is the product of force and time which equals change in momentum
(2) The notion of impulse is useful when a large force acts for a short time to produce a measurable change in momentum
(3) There must be an appreciable change in the position of the body during the action of the impulsive force
(4) SI unit of impulse is kg m s^{-1}

41. Select the correct option about the friction between the two bodies.
(1) Static friction is always greater than the kinetic friction
(2) Limiting friction is always greater than the kinetic friction
(3) Limiting friction is not always greater than the kinetic friction
(4) The direction of friction is always opposite to motion

42. The tension in the string PQ as shown in the figure is (take $g = 10\ m/s^2$)

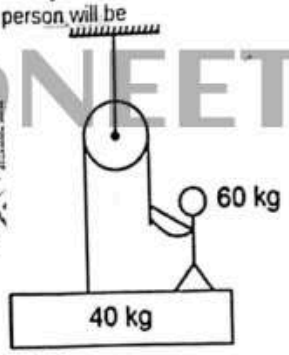


(1) 20 N
(2) 40 N
(3) $20\sqrt{2}$ N
(4) $40\sqrt{2}$ N

43. A block of mass 2 kg is to be kept at rest against a rough vertical wall ($\mu = 0.4$) by applying a horizontal force F. The minimum value of F is ($g = 10\ m/s^2$)
(1) 20 N
(2) 40 N
(3) 50 N
(4) 45 N

44. A force (in N) vector applied on a mass is represented as $\vec{F} = 6\hat{i} - 8\hat{j} + 10\hat{k}$ and it accelerates with 1 m/s^2. What will be the mass of the body?
(1) 10 kg
(2) 20 kg
(3) $2\sqrt{10}$ kg
(4) $10\sqrt{2}$ kg

45. A person of mass 60 kg is standing on a platform attached to rope and pulley as shown. If mass of platform is 40 kg and the system is at rest, then normal reaction on the feet of person will be



(1) 20g
(2) 15g
(3) 10g
(4) 5g

CHEMISTRY

46. Electronic configuration of elements A, B, C and D are as follows
A: $1s^2 2s^2 2p^4$ B: $1s^2 2s^2 2p^5$
C: $1s^2 2s^2 2p^6 3s^2 3p^5$ D: $1s^2 2s^2 2p^3$
The most electronegative element from the given elements is
(1) A
(2) B
(3) C
(4) D

47. The IUPAC name of element with atomic number 104 is
(1) Unnilennium
(2) Unnilquadium
(3) Ununoctium
(4) Unnilseptium

48. Which of the following has a positive value of electron gain enthalpy?
(1) Hydrogen
(2) Sulphur
(3) Neon
(4) Oxygen

49. Which of the following is a set of isoelectronic species?
(1) CO, N_2, CN^-
(2) N_2, O_2^+, NO
(3) CO, CN^-, O_2^{-2}
(4) NO_2, N_2O, O_3

50. Among the following, the element which has largest atomic size is
(1) O
(2) P
(3) N
(4) Cl

51. Lithium is diagonally related with
(1) B
(2) Al
(3) Be
(4) Mg

52. Which of the following is/are not Dobereiner's triad?
(a) Li, Na, K
(b) Cu, Ag, Zn
(c) Fe, Co, Ni
(d) Ca, Sr, Ba
Correct answer is
(1) (a) & (b) only
(2) (a) & (d) only
(3) (b) & (c) only
(4) (a), (b), (c) & (d)

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53. The first ionisation potential of Na is 5.1 eV. The value of electron gain enthalpy of Na^+ will be
(1) –5.1 eV (2) –10.2 eV
(3) +2.55 eV (4) –2.55 eV

54. Match the IUPAC names of the elements given in column I with their IUPAC symbols given in column II.

Column I	Column II
a. Unnilpentium	(i) Hs
b. Unnilhexium	(ii) Bh
c. Unnilseptium	(iii) Sg
d. Unniloctium	(iv) Db

Identify the correct match.
(1) a(iv), b(iii), c(ii), d(i) (2) a(iv), b(i), c(ii), d(iii)
(3) a(ii), b(i), c(iv), d(iii) (4) a(ii), b(iii), c(iv), d(i)

55. Ion which has the largest size is
(1) N^{3-} (2) O^{2-}
(3) Na^+ (4) Mg^{2+}

56. Which of the following is a representative element?
(1) Zn (2) As
(3) Cu (4) Fe

57. If the atomic number of an element is 60, it will be placed in the periodic table in the
(1) 4th group, 7th period (2) 4th group, 6th period
(3) 3rd group, 7th period (4) 3rd group, 6th period

58. A 100 watt bulb emits monochromatic light of wavelength 400 nm. The number of photons emitted per second by the bulb is [$h = 6.6 \times 10^{-34}$ Js]
(1) 1×10^{15} (2) 2×10^{20}
(3) 2×10^{10} (4) 8×10^{20}

59. Based on the Bohr's model match List I with the list II

	List I		List II
a.	Kinetic energy of the electron in the nth orbit	(i)	$\propto n$
b.	Angular momentum of the electron in the nth orbit	(ii)	$\propto n^2$
c.	Radius of nth orbit	(iii)	$\propto n^{-1}$
d.	Velocity of electron in the nth orbit	(iv)	$\propto n^{-2}$

Choose the correct option
(1) a(iv), b(i), c(ii), d(iii) (2) a(ii), b(i), c(iv), d(iii)
(3) a(iii), b(i), c(ii), d(iv) (4) a(iv), b(iii), c(ii), d(i)

60. Which of the following sets of quantum numbers may be correct for an electron in 4f orbital?
(1) n = 4, l = 3, m_l = –2, s = 0
(2) n = 4, l = 3, m_l = +4, s = $-\frac{1}{2}$
(3) n = 4, l = 3, m_l = +1, s = $+\frac{1}{2}$
(4) n = 4, l = 4, m_l = –1, s = $+\frac{1}{2}$

61. A 3.3 mg ball is moving with a speed of 200 m/s. The de Broglie wavelength associated with it will be ($h = 6.6 \times 10^{-34}$ Js)
(1) 10^{-27} m (2) 10^{-30} m
(3) 10^{-24} m (4) 10^{-33} m

62. Maximum number of electron(s) which can be accommodated in a p-orbital having anticlockwise spin is
(1) 6 (2) 3
(3) 2 (4) 1

63. Which among the following is incorrect observation during the photoelectric effect?
(1) No time lag between the striking of light and ejection of electrons
(2) The number of electrons ejected does not depend upon the frequency of light used
(3) The kinetic energy of the photoelectrons depends upon the frequency of light used
(4) If the frequency of light used is lesser than the threshold frequency only then electrons are ejected from metal

64. Ratio of energy of electron of 1st orbit of hydrogen, 2nd orbit of $He^{\oplus}$ ion and 3rd orbit of Li^{2+} ion will be
(1) 1 : 2 : 3 (2) 1 : 1 : 1
(3) 1 : 4 : 9 (4) 3 : 2 : 1

65. Radial nodes present in 3s and 3p-orbitals are respectively
(1) 0, 2 (2) 2, 1
(3) 1, 1 (4) 2, 2

66. The uncertainty in momentum of an electron is 1×10^{-5} kg-m/s. The uncertainty in its position will be ($h = 6.62 \times 10^{-34}$ kg-m^2/s)
(1) 5.27×10^{-30} m (2) 1.05×10^{-26} m
(3) 1.05×10^{-28} m (4) 5.25×10^{-28} m

67. In Bohr's model of hydrogen atom, if atomic radius of the first orbit is x, then the radius of the 3rd orbit will be
(1) $\frac{x}{3}$ (2) 3x
(3) 9x (4) 6x

68. The wavelength of an electromagnetic radiation is 400 nm. Thus, the wave number will be equal to
(1) 2.5×10^4 cm^{-1} (2) 2.5×10^3 cm^{-1}
(3) 2.5×10^5 cm^{-1} (4) 2.5×10^2 cm^{-1}

69. What will be the ratio of the wavelength of the first line to that of the second line of Paschen series of H atom?
(1) 256 : 175 (2) 175 : 256
(3) 15 : 16 (4) 24 : 27

70. How many electrons in Ca atom can have m_l = 0?
(1) 2 (2) 12
(3) 10 (4) 20

71. In Lyman series the longest wavelength for hydrogen atom will be (R_H = Rydberg constant)
(1) R_H (2) $\frac{1}{R_H}$
(3) $\frac{4}{3R_H}$ (4) $\frac{36}{5R_H}$

72. Match List I with List II :

	List I (Quantum Numbers)			List II (Orbital)
	'n'	'l'		
A.	2	1	I.	3d
B.	4	0	II.	2p
C.	5	3	III.	4s
D.	3	2	IV.	5f

Choose the **correct** answer from the options given below.
(1) A-IV, B-II, C-III, D-I (2) A-II, B-III, C-I, D-IV
(3) A-II, B-III, C-IV, D-I (4) A-I, B-II, C-III, D-IV

73. In H-atom spectrum electron jumps from 5th excited state to 1st excited state then total number of spectral lines, number of lines in Lyman series and Paschen series respectively are
(1) 10, 4, 3 (2) 15, 0, 4
(3) 15, 4, 5 (4) 10, 0, 3

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74. Given below are two statements
Statement I: The shapes of d_{xy}, d_{yz}, d_{xz} and d_{z^2} orbitals are similar to each other
Statement II: All five 3d orbitals are equivalent in energy in free state.
In the light of above statements, choose the correct answer from the options given below
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

75. Match the following columns:

Column I (Quantity)	Column II (Number of atoms)
a. 1.8 g of water	(i) 0.6 N_A
b. 3.1 g of phosphorus (P_4)	(ii) 0.05 N_A
c. 1.6 g of sulphur (S_8)	(iii) 0.3 N_A
d. 9.2 g of nitrogen dioxide	(iv) 0.1 N_A

Choose the option containing most appropriate match:
(1) a(iii); b(ii); c(i); d(iv) (2) a(iv); b(i); c(ii); d(iii)
(3) a(iii); b(iv); c(ii); d(i) (4) a(iv); b(iii); c(ii); d(i)

76. Mass of 60% (w/w) H_2SO_4 solution required to react completely with 20 g $CaCO_3$ (50% pure) would be
(1) 14 g (2) 28 g
(3) 13.45 g (4) 16.33 g

77. During the Ostwald's process, if 34 g of NH_3 gas reacts with 44.8 L of O_2(g) at STP, then consider the following statements.
$4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g)$
(a) O_2 is limiting reagent
(b) 48.0 g of NO(g) is formed
(c) 143.36 L gaseous product present at the end of reaction at STP
(d) 53.76 L of H_2O (g) is formed at STP
The correct statements are
(1) (a) and (d) only (2) (b), (c) and (d) only
(3) (a), (b) and (d) only (4) (b) and (c) only

78. Given below are two statements.
Statement I: If an aqueous solution of 12.6 g oxalic acid dihydrate is made upto 1000 mL, then the volume of 0.1 N NaOH required to completely neutralize 10 mL of this oxalic acid dihydrate solution will be 20 mL.
Statement II: A gaseous mixture contains nitrogen gas and oxygen gas in the ratio of 1 : 2 by weight. Therefore, the ratio of moles of nitrogen gas to oxygen gas will be 4 : 7 (consider gases are not reacting with each other).
In the light of above statements, choose the **correct** answer from the option given below.
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

79. Match the following columns:

Column I (Compound)	Column II (Percentage of oxygen)
a. Sulphuric acid	(i) 28.5%
b. Ammonium sulphate	(ii) 50%
c. Sulphur dioxide	(iii) 48.48%
d. Calcium oxide	(iv) 65.3%

Choose the correct option containing most appropriate match.
(1) a(iii); b(i); c(iv); d(ii) (2) a(iii); b(i); c(ii); d(iv)
(3) a(iv); b(iii); c(i); d(ii) (4) a(iv); b(iii); c(ii); d(i)

80. Match prefixes given in List-I with multiple given in List-II

List-I	List-II
a. mega	(i) 10^{18}
b. giga	(ii) 10^{6}
c. exa	(iii) 10^{12}
d. tera	(iv) 10^{9}

The correct match is
(1) a(iii), b(ii), c(i), d(iv) (2) a(ii), b(iv), c(iii), d(i)
(3) a(iv), b(ii), c(i), d(iii) (4) a(ii), b(iv), c(i), d(iii)

81. Which of the following acts as limiting reagent if 3 mole of A, 1 mole of B, 2 mole of C and 3 mole of D react according to given equation?
$2A + 4B + 5C + D \rightarrow 4x + y + 3z$
(1) A (2) B
(3) C (4) D

82. If mass percentage of glucose in aqueous solution is 36% then the molality of the solution is
(1) 2.250 (2) 3.125
(3) 5.125 (4) 4.612

83. If 0.3 mole of $BaCl_2$ is mixed with 0.4 moles of Na_3PO_4, then maximum number of moles of $Ba_3(PO_4)_2$ and NaCl formed are respectively
(1) 0.2 and 1.6 (2) 0.3 and 1.8
(3) 0.1 and 0.6 (4) 0.4 and 2.4

84. 50 g of an impure calcium carbonate sample decomposes on heating to give carbon dioxide and 22.4 g calcium oxide. The percentage purity of calcium carbonate in the sample is
(1) 60% (2) 80%
(3) 90% (4) 70%

85. 34.2 g of aluminium sulphate is dissolved in enough water to make 500 mL of solution. If aluminium sulphate dissociates completely, then the molar concentration of Al^{3+} ions and SO_4^{2-} ions respectively are
(1) 0.4 M & 0.4 M (2) 0.2 M & 0.4 M
(3) 0.4 M & 0.6 M (4) 0.3 M & 0.6 M

86. 4 g of Mg is reacted with 8 g of O_2. Mole of MgO produced in the reaction is
(1) $\frac{1}{4}$ (2) $\frac{1}{5}$
(3) $\frac{1}{6}$ (4) $\frac{1}{3}$

87. The mass of urea required in making 2.5 kg of 0.25 molal aqueous solution is
(1) 42 g (2) 32 g
(3) 37 g (4) 45 g

88. An organic compound contains carbon, hydrogen and oxygen. Its elemental analysis gave C = 40% and H = 6.67%. The empirical formula of the compound would be
(1) HCO (2) H_2CO
(3) HC_2O (4) H_2C_2O

89. Mole fraction of the solute in 0.5 molal aqueous solution is nearly

(1) $\frac{1}{150}$ (2) $\frac{1}{56}$

(3) $\frac{1}{112}$ (4) $\frac{1}{25}$

90. The crystalline salt $Na_2CO_3.xH_2O$ on heating loses 63% of its mass and becomes anhydrous. The formula of the compound is

(1) $Na_2CO_3.2H_2O$ (2) $Na_2CO_3.10H_2O$

(3) $Na_2CO_3.5H_2O$ (4) $Na_2CO_3.8H_2O$

BOTANY

91. Read the following statements of Assertion and Reason.

Assertion : Endoplasmic reticulum principally performs the function of packaging materials, to be delivered either to the intra-cellular targets or secreted outside the cell.

Reason : Endoplasmic reticulum passes materials from Golgi bodies to outside the cell membrane.

In the light of above statements choose the **correct** answer from the options given below.

(1) Both Assertion and Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion and Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

92. Read the following statements:

(A) Different cells have diversity in size, shape and functions.

(B) The organisation of the prokaryotic cell is fundamentally similar even though prokaryotes exhibit a wide variety of shapes and functions.

(C) Nerve cells are the longest, unbranched cells of human body.

(D) The largest isolated single cell is the egg of an ostrich.

(E) Plasmid DNA is used to monitor bacterial transformation with foreign DNA.

The **incorrect** one(s) is/are

(1) Only (D) and (E) (2) Only (C)

(3) Only (B) (4) Only (A) and (B)

93. Read the following statements:

(a) Many membrane bound minute vesicles that contain various enzymes, are present in both plant and animal cells.

(b) A single human cell has approximately two meter long thread of DNA which is distributed among its forty six chromosomes.

(c) Svedberg's unit is indirectly a measure of density and size.

(d) The central part of the proximal region of the centriole is also proteinaceous and it is known as axoneme

(e) A few chromosomes have staining secondary constrictions at a constant location, known as satellite.

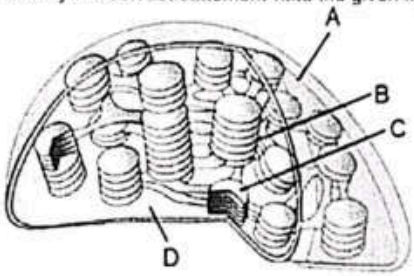
The **incorrect** ones are

(1) (a) and (d) (2) (a), (b) and (c)

(3) (d) and (e) only (4) (b), (c) and (d)

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94. Identify the **correct** statement w.r.t. the given figure.



A. It is more permeable in comparison to the inner membrane.

B. These are stacked flat membranous tubules arranged like piles of coins which help in CO_2 fixation.

C. These are organised flattened membranous sacs present in the stroma.

D. It has enzymes for synthesis of sugar.

(1) A and B only (2) B and C only

(3) A, C and D (4) B, C and D

95. Which of the following statements regarding mitochondria is incorrect?

(1) Outer membrane is permeable to monomers of carbohydrates, fats and proteins.

(2) Enzymes of electron transport are embedded in outer membrane.

(3) Inner membrane is convoluted with infoldings.

(4) Mitochondrial matrix contains single circular DNA molecule and ribosomes.

96. Match column I with column II and choose the **correct** option.

	Column I		Column II
a.	Metacentric	(i)	Centromere of chromosome is situated close to one of its end.
b.	Acrocentric	(ii)	They appear "L-shaped" during anaphase
c.	Sub metacentric	(iii)	Chromosome appears to have a single arm
d.	Telocentric	(iv)	It is also called isobrachial chromosome

(1) a(ii), b(iv), c(i), d(iii) (2) a(iv), b(i), c(ii), d(iii)

(3) a(iii), b(i), c(iv), d(ii) (4) a(iv), b(iii), c(i), d(ii)

97. Select the **incorrect** statement w.r.t. lysosome.

(1) This organelle is rich in nearly all types of hydrolytic enzymes

(2) Its enzymes are optimally active at acidic pH

(3) It is involved in the biosynthesis of peroxides

(4) It is formed by the process of packaging in golgi apparatus

98. The site for ribosomal RNA synthesis in eukaryotes

(1) Is a spherical structure found in the nucleoplasm

(2) Is bounded by two membranes

(3) Is absent in cells which are involved in protein synthesis

(4) Has large number of ribosomes

99. Identify the **incorrect** match for plasma membrane.
(1) Tails of lipids – Hydrophobic in nature
(2) Integral proteins – May partially buried in membrane
(3) Proteins – Show flip-flop movement
(4) Heads of lipids – Interact with water

100. Select the **incorrectly** matched pair.
(1) Robert Brown – Discovered the nucleus of a cell
(2) Robert Hooke – Discovered the cell from a thin slice of cork
(3) A.V. Leeuwenhoek – First person to observe and describe a living cell.
(4) Matthias Schleiden – A British zoologist, who studied different types of animal cells.

101. Eukaryotic flagella and cilia are similar to each other as both
a. Have '9+2' organisation of microtubules
b. Are surrounded by membrane
c. Help in locomotion
d. Give cartwheel appearance with no microtubules in the centre.
The **correct** one(s) is/are
(1) Only a and b (2) Only c
(3) Only a and d (4) Only a, b and c

102. Select the statement which is **not** true for microbodies.
(1) These are single membrane bound structures
(2) They contain various enzymes
(3) They are present in both plants and animals
(4) They are present in both nucleoplasm and cytoplasm

103. The organelle important for osmoregulation and excretion in *Amoeba*
(1) Performs the function of packaging materials
(2) Contains 70S ribosomes
(3) Consists of cisternae that are concentrically arranged near the nucleus
(4) Is bound by a single membrane

104. The ribosomes of polysome translate the mRNA into proteins. This process occurs in
(a) Mitochondria
(b) Cytosol
(c) Nucleoplasm
(d) Chloroplast
The **correct** one(s) is/are
(1) (b) only (2) (a), (b) and (d)
(3) (b) and (c) (4) (c) only

105. A bacterium attaches to the host cell with the help of bristle-like structure is called
(1) Pili (2) Flagella
(3) Fimbriae (4) Cell wall

106. The type of plastid which is **not** involved in storage of food, is
(1) Chloroplast (2) Amyloplast
(3) Chromoplast (4) Elaioplast

107. Diplotene stage is characterised by
(a) Dissolution of synaptonemal complex.
(b) Terminalisation of chiasmata.
(c) Complete disappearance of nucleolus.
(d) Appearance of chiasmata.
(e) Disintegration of nuclear envelope.
Choose the **correct** option.
(1) (a), (b) & (c) (2) (c), (d) & (e)
(3) (a) & (d) (4) All except (a)

108. Match the following columns and select the **correct** option.

	Column-I		Column-II
a.	G_1 phase	(i)	Cells in this stage remain metabolically active but do not proliferate
b.	Prophase	(ii)	Interval between most active phase of cell cycle and initiation of DNA replication
c.	Telophase	(iii)	Chromosomal material becomes untangled during the process of chromatin condensation
d.	G_0 phase	(iv)	Chromosome cluster at opposite spindle poles and their identity is lost as discrete elements

(1) a(i), b(ii), c(iv), d(iii) (2) a(ii), b(iii), c(iv), d(i)
(3) a(ii), b(i), c(iii), d(iv) (4) a(i), b(iv), c(iii), d(ii)

109. Meiosis involves
(1) Two sequential cycles of DNA replication
(2) Only one cycle of nuclear division
(3) One cycle of DNA replication
(4) Two cycles of nuclear division and only one cycle of cytoplasm division

110. Match the column I with column II.

	Column I		Column II
a.	Bivalent	(i)	Multinucleated condition in some organisms
b.	Chiasmata	(ii)	More clearly visible as tetrads at the third sub-stage of prophase I
c.	Recombination nodules	(iii)	Becomes visible during the fourth sub-stage of prophase I
d.	Syncytium	(iv)	Are formed at the sites where crossing over occurs

Choose the **correct** option.
(1) a(iv), b(i), c(iii), d(ii) (2) a(iii), b(iv), c(ii), d(i)
(3) a(iv), b(ii), c(i), d(iii) (4) a(ii), b(iii), c(iv), d(i)

111. In an onion root tip cell, there are 16 chromosomes. How many chromosomes will be present in the following phases of meristematic cells of that root tip?
(a) In G_1 phase
(b) After S-phase
(c) After M phase
Choose the **correct** answer from the options given below.
(1) (a)-16, (b)-16, (c)-16 (2) (a)-16, (b)-32, (c)-32
(3) (a)-32, (b)-16, (c)-32 (4) (a)-32, (b)-32, (c)-32

112. Select the **odd** one w.r.t. significance of mitosis.
(1) Growth of multicellular organisms
(2) Maintenance of cell size
(3) Maintenance of chromosome number
(4) Introduction of variations

113. Mark the wrong statement w.r.t intrameiotic interphase.
(1) Chromosomes number is exactly half to the meiocyte
(2) It is short lived
(3) Duplication of DNA takes place
(4) Centrioles pair duplicate in animal cells

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114. Given below are the two statements: one is labelled as assertion (A) and the other is labelled as reason (R).
Assertion (A): Telophase is the final stage of M-phase.
Reason (R): In telophase, chromatin moves towards the two poles, its elongation starts and nucleolus reappears.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is the correct explanation of (A)
(3) Both (A) and (R) are false
(4) (A) is true but (R) is false

115. Disc-shaped proteinaceous structure associated with centromere is called
(1) Kinetochore
(2) Secondary constriction
(3) Primary as well as secondary constriction
(4) Nucleolus organizer region

116. During meiosis, separation of sister chromatids occurs in
(1) Anaphase-I (2) Prophase-II
(3) Metaphase-II (4) Anaphase-II

117. Resting phase of cell cycle is
(1) Prophase (2) Interphase
(3) M-phase (4) Only S-phase

118. If all the daughter cells produced are capable to divide again then, how many generations are required to produce 128 cells from such a cell through mitosis?
(1) 127 (2) 128
(3) 6 (4) 7

119. Read the following statements and select the correct option.
Statement A: Taxonomic categories are distinct biological entities and not merely morphological aggregates.
Statement B: Classification of organisms into various categories does not help in identifying similarities and dissimilarities among different organisms.
(1) Only A is incorrect
(2) Only B is incorrect
(3) Both A and B are correct
(4) Both A and B are incorrect

120. *Panthera leo* and *Canis familiaris* are
(1) Placed in same kingdom and same genus
(2) Placed in different orders and different families
(3) Placed in same kingdom and different families
(4) Placed in different kingdom and different orders

121. Arrange the following taxonomic categories in the ascending order w.r.t. wheat and select the **correct** option.
(A) Poales
(B) *Triticum*
(C) Monocotyledonae
(D) Angiospermae
(E) Poaceae
The correct sequence is as follows
(1) B, E, D, C, A (2) E, B, D, C, A
(3) B, E, C, A, D (4) B, E, A, C, D

122. Choose the **incorrect** match.
(1) Family – A group of related genera
(2) Order – A group of related families
(3) Class – A group of related orders
(4) Division – A group of related phyla

123. Modern taxonomic studies are based on all, **except**
(1) External and internal features
(2) Ecological informations
(3) Developmental processes
(4) Phylogeny

124. Out of the following taxa which one represents the highest rank in taxonomic hierarchy?
(1) Poales (2) Anacardiaceae
(3) Monocotyledonae (4) Polymoniales

125. Binomial epithet in a scientific name includes
(1) Generic name and family name only
(2) Generic name and author citation only
(3) Specific epithet and author citation only
(4) Generic name, specific epithet and author citation

126. Select the **mismatched** pair w.r.t. category of bacteria and their shapes
(1) Coccus – Cylindrical
(2) Bacillus – Rod shaped
(3) Vibrium – Comma shaped
(4) Spirillum – Spiral

127. Eubacteria and archaebacteria were put in which of the following kingdoms by R.H. Whittaker?
(1) Protista (2) Monera
(3) Plantae (4) Fungi

128. The three domains of living organisms - Archaea, Bacteria and Eukarya have six kingdoms. Domain Eukarya includes how many kingdom(s)?
(1) 2 (2) 3
(3) 1 (4) 4

129. Archaebacteria differ from eubacteria as the former
(1) Have a different cell wall structure
(2) Have symbiotic association with eukaryotes
(3) Do not have primitive nucleus
(4) Do not have histone proteins

130. Blue green algae
(1) Are eukaryotes
(2) Have prominent nucleus
(3) Are also known as cyanobacteria
(4) Are grouped under Chlorophyceae

131. Neutral solutes moving across plasma membrane do not
(1) Diffuse along concentration gradient
(2) Move from their higher concentration to lower concentration
(3) Show passive transport
(4) Require energy for their movement from their higher to lower concentration

132. A specialised membranous structure in some prokaryotic cells which contain photosynthetic pigments is
(1) Mesosome (2) Cristae
(3) Chromatophore (4) Endoplasmic reticulum

133. A bacterium divides amitotically in 20 minutes. A suspension culture of these bacteria containing 10^3 cells/ml is grown for 160 minutes. Find out cell concentration per ml in 160 minutes.
(1) 8×10^3 (2) 10^{768}
(3) 256×10^3 (4) 160×10^3

134. Duplication of the Golgi apparatus occurs during the
(1) G_1-phase (2) S-phase
(3) G_2-phase (4) Prophase

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135. Read the following statements

a. In contrast to mitosis, meiosis occurs in the diploid cells, which are destined to form gametes.

b. At the end of anaphase I, each pole receives half the chromosome number of the parent cell.

c. Mitosis and meiosis I are the equational division in which the chromosome number of the parent is conserved in the daughter cell.

d. Interkinesis is the only phase that marks the phase of chromosome duplication.

e. The actual period of cell division is further subdivided into G_1, S and G_2.

Select the **correct** ones:

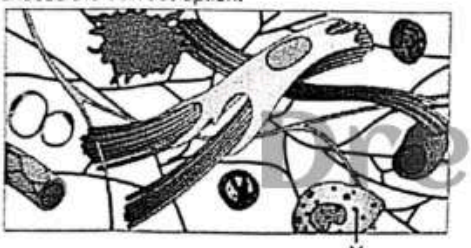
(1) c, d and e only
(2) a only
(3) a and b only
(4) a, c and e only

ZOOLOGY

136. The epithelium which is made of a single thin layer of flattened cells with irregular boundaries is found in

(1) PCT of nephron
(2) Walls of blood vessels
(3) Inner surface of hollow organs like bronchioles
(4) Stomach and intestine

137. Identify the structure marked X in below given figure and choose the **correct** option.



(1) Mast cell
(2) Collagen
(3) Fibroblast
(4) Goblet cell

138. Which type of junctions in the epithelial tissues facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells for rapid transfer of ions or small molecules?

(1) Tight junctions
(2) Adhering junctions
(3) Gap junctions
(4) Desmosomes

139. Muscle fibres which taper at both ends are _____ in structure.

Choose the option which fills the blank **correctly.**

(1) Skeletal
(2) Cylindrical
(3) Unbranched
(4) Multinucleated

140. All of the following structures have cartilage, **except**

(1) In the tip of nose
(2) Between adjacent bones of the vertebral column, limbs in adult
(3) Nasal septum, hands in adult, thoracic cage
(4) Heart, Kidney, Capillaries

141. Complete the analogy

Multicellular gland : Saliva : : Unicellular gland : __

(1) Mucus
(2) Milk
(3) Oil
(4) Earwax

142. Read the features listed below.

(a) Fibres and fibroblasts are compactly packed

(b) Collagen fibres are present in rows between many parallel bundles of fibres

(c) Present in structure which connects skeletal muscles to bones

Select the correct option that represents the above mentioned features.

(1) Neural tissue
(2) Areolar tissue
(3) Dense regular connective tissues
(4) Dense irregular connective tissue

143. The type of epithelium present in the inner lining of ducts of salivary gland, is also present in the

(1) Walls of blood vessels
(2) Lining of stomach and intestine
(3) Inner surface of hollow organs like bronchioles
(4) Moist surface of buccal cavity

144. In humans, the simple squamous epithelium is usually not found in the structure that

(1) Allows flow of blood in a fixed route
(2) Assists in exchange of gases between blood and atmospheric air
(3) Forms the part of filtration membrane that participates in ultrafiltration
(4) Acts as the site of fertilisation

145. Division of labour by assigning different functions to cells forming different types of tissue was first observed in the phylum

(1) Platyhelminthes
(2) Cnidaria
(3) Aschelminthes
(4) Porifera

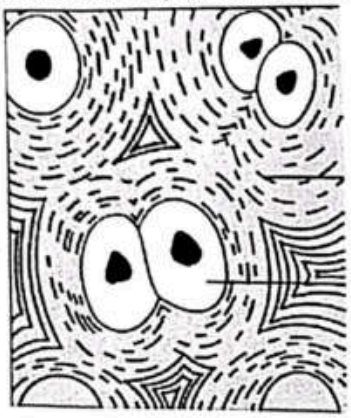
146. Read the following statements and choose the **correct** option.

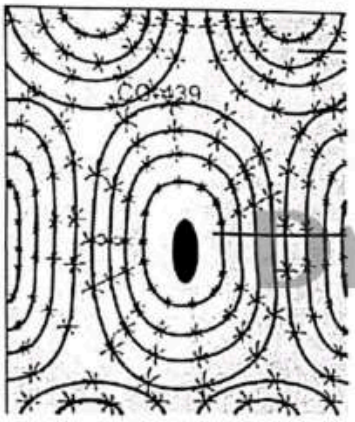
Statement A: When a neuron is suitably stimulated, an electrical disturbance is generated.

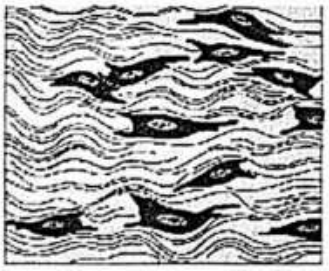
Statement B: Arrival of the disturbance at the neuron endings always stimulate adjacent neurons.

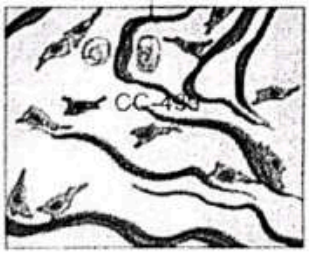
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

147. Read the following:
(a) Ground substance is hard and non-pliable.
(b) Rich in collagen fibres and calcium salts.
Identify the tissue on the basis of above characteristics and choose the **correct** option from the given diagrams.

(1)


(2)


(3)


(4)


148. Choose the **incorrect** statement w.r.t. neuroglial cells of the nervous tissue.
(1) These are the cells that provide insulation to the nerve cells
(2) These are non-excitable cells and provide protection to the neurons
(3) They form the majority of the neural tissue
(4) They are the unit of neural system

149. Function of microvilli of epithelial cells is to
(1) Protect the cell
(2) Increase the surface area for absorption
(3) Move particles or mucus in a specific direction
(4) Engulf the foreign matter

150. The nasal chamber directly opens into
(1) Pharynx (2) Trachea
(3) Bronchi (4) Alveoli

151. Ribozymes are _____ that behave like enzymes.
Fill in the blank by selecting the correct option.
(1) Proteins (2) Ribonucleic acids
(3) Oligosaccharide (4) Simple lipids

152. Match column I with column II and select the **correct** option.

	Column I		Column II
a.	Collagen	(i)	Most abundant protein in the whole of the biosphere
b.	RuBisCO	(ii)	Most abundant protein in animal world
c.	Cellulose	(iii)	Hormone that increases blood glucose levels
d.	Glucagon	(iv)	Homopolymer which is also a secondary metabolite

(1) a(ii), b(iv), c(iii), d(i) (2) a(iii), b(iv), c(ii), d(i)
(3) a(i), b(ii), c(iii), d(iv) (4) a(ii), b(i), c(iv), d(iii)

153. Choose the option which represents the correct decreasing order of cellular components according to their percentage of the total cellular mass.
(1) Water > Proteins > Nucleic acids > Carbohydrates > Lipids > Ions
(2) Nucleic acids > Proteins > Lipids > Carbohydrates > Ions > Water
(3) Proteins > Nucleic acids > Ions > Carbohydrates > Water > Lipids
(4) Ions > Lipids > Carbohydrates > Nucleic acids > Proteins > Water

154. A double stranded DNA contains 20% of nitrogenous bases as guanine. What will be the expected percentage of other three nitrogenous bases i.e. cytosine, adenine and thymine represented as C, A and T?
(1) C - 20%, A - 30%, T - 30%
(2) C - 30%, A - 20%, T - 30%
(3) A - 40%, T - 20%, C - 20%
(4) A - 20%, T - 20%, C - 40%

155. In an enzyme-catalysed reaction, the presence of certain molecules increases the apparent K_m while leaving the V_{max} unchanged. The above said fact is observed during
(1) Non-competitive inhibition
(2) Competitive inhibition
(3) Endothermic reactions only
(4) Enzyme denaturation

156. The retentate obtained after straining the slurry made by grinding a living tissue in trichloroacetic acid contains which set of compounds?
(1) Proteins, minerals, lipids
(2) Phosphate ions, sulphate ions, magnesium ions
(3) Phosphate ions, sulphate ions, proteins
(4) Proteins, nucleic acids, polysaccharides

157. How many statements is/are **incorrect** w.r.t. the enzymes?
(a) Enzymes are divided into 6 classes, each with 4-13 subclasses and named accordingly by a 4-digit number.
(b) Ligases catalyse the joining of C – O, C – S, C – N, P – O, *etc*, bonds.
(c) Hydrolases catalyse the hydrolysis of ester, ether, peptide and glycosidic bonds.
(d) Catalytic activity is lost permanently when the enzymes are exposed to low temperature.
(1) Two (2) Three
(3) Four (4) One

158. Select the **incorrect** match.

(1)	GLUT-4	–	Enables glucose transport into cells
(2)	Receptor	–	Sensory reception (smell, taste, *etc.*)
(3)	Antibody	–	Does not fight against infectious agents
(4)	Trypsin	–	Proteolytic enzyme

(1) (1) (2) (2)
(3) (3) (4) (4)

159. Which of the following are **correct** with reference to amino acids?
a. Amphoteric in nature.
b. Are substituted methane molecules.
c. Serine is an alcoholic amino acid.
d. Amino acids never exist in zwitterionic form at any stage in living cells.
e. A dipeptide comprises of two peptide bonds between amino acids.
(1) a, b, c (2) a, d, e
(3) b, c, d (4) a, c, d, e

160. How many of the compounds given in the box below possess heterocyclic ring(s) in their structure?

Ribose, Serine, Glycerol, Adenylic acid, Uracil, Cholesterol

Select the correct option.
(1) Four (2) Three
(3) Two (4) Five

161. The major component which is present in the plant cell wall, does not give colour while performing iodine test, because it is/does
(1) A homopolymer
(2) A polysaccharide
(3) Not contain complex helices in its structure
(4) Not have a reducing end

162. Select the **correct** set of molecules that do not contain a glycosidic bond in their structures and are heteropolymers.
(1) Cytidylic acid and guanosine
(2) RNA and DNA
(3) Glycogen and inulin
(4) Insulin and GLUT-4

163. Match column I with column II.

Column I	Column II
a. Acidic amino acid	(i) Valine
b. Basic amino acid	(ii) Glutamic acid
c. Neutral amino acid	(iii) Lysine
d. Aromatic amino acid	(iv) Phenylalanine

Select the **correct** option.
(1) a-(i), b-(ii), c-(iii), d-(iv) (2) a-(iv), b-(i), c-(ii), d-(iii)
(3) a-(ii), b-(iii), c-(i), d-(iv) (4) a-(iv), b-(iii), c-(ii), d-(i)

164. DNA resembles RNA as both
(1) Are the genetic material in humans
(2) Have similar sugars
(3) Contain similar pyrimidine bases
(4) Are polymer of nucleotides

165. In a nucleotide, the bond present between the phosphate and hydroxyl group of a sugar is
(1) Peptide bond (2) Ester bond
(3) Glycosidic bond (4) Hydrogen bond

166. Read the following statements on lipids and find out correct set of statements:
(a) Lecithin found in the plasma membrane is a glycolipid.
(b) Saturated fatty acids possess one or more C = C bonds.
(c) Gingelly oil has lower melting point and remains as oil in winter.
(d) Lipids are generally insoluble in water but soluble in some organic solvents.
(e) When a fatty acid is esterified with glycerol, monoglyceride is formed.
Choose the correct answer from the options given below:
(1) (c), (d) and (e) only (2) (a), (b) and (d) only
(3) (a), (b) and (c) only (4) (a), (d) and (e) only

167. A co-factor which forms coordination bonds with side chains at the active site of enzyme carboxypeptidase and with its substrate is
(1) Calcium (2) Zinc
(3) Haem (4) Vitamin

168. The volume of air that will remain in the lungs after a normal expiration is equivalent to
(1) TLC (2) ERV + RV
(3) TV + ERV (4) IC + ERV

169. During inspiration, the volume of thoracic cavity
(1) Increases in antero-posterior axis
(2) Decreases in dorso-ventral axis
(3) Decreases in antero-posterior axis
(4) Decreases in cephalo-caudal axis

170. Cellular respiration includes
(1) Exchange of gases across alveolar membrane
(2) Pulmonary ventilation
(3) Breakdown of glucose and release of CO_2
(4) Diffusion of gases between blood and tissue

171. Match the items given in Column-I with those in Column-II w.r.t. animals and their respiratory organs and choose the **correct** option.

	Column-I		Column-II
a.	*Labeo*	(i)	Lungs
b.	*Pheretima*	(ii)	Tracheal tubes
c.	*Periplaneta*	(iii)	Moist cuticle
d.	*Pteropus*	(iv)	Gills

(1) a(i), b(ii), c(iii), d(iv) (2) a(iv), b(iii), c(ii), d(i)
(3) a(iii), b(iv), c(i), d(ii) (4) a(ii), b(i), c(iii), d(iv)

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172. The first step of respiration is
(1) Transport of gases from lungs to blood
(2) Transport of gases from blood to tissues
(3) Pulmonary ventilation
(4) Catabolism of food substances to release energy within cells

173. Choose the **correct** option w.r.t. the 'sound box' present in humans.
(1) Present in the midline of the neck posterior to the oesophagus
(2) Its lower most part has an opening, called glottis
(3) Protects the lower respiratory tract from aspirating food into the trachea while swallowing food
(4) Not a part of the conducting zone of the respiratory passage

174. The thoracic chamber is formed dorsally and ventrally respectively by
(1) Vertebral column and Sternum
(2) Sternum and Vertebral column
(3) Ribs and Sternum
(4) Diaphragm and Clavicle

175. Read the following given statements and choose the **correct** option.
Statement-A: Cutaneous respiration occurs through vascularised bags called lungs.
Statement-B: Pulmonary respiration occurs through the moist skin.
(1) Only statement A is incorrect
(2) Only statement B is incorrect
(3) Both statements A and B are incorrect
(4) Both statements A and B are correct

176. Read the following statements w.r.t humans:
Statement (A): Breathing is a chemical process, while cellular respiration is purely mechanical.
Statement (B): Breathing ensures oxygen intake and carbon dioxide removal while cellular respiration is utilisation of O_2 by the cells for catabolic reactions and resultant release of CO_2.
Choose the **correct** option.
(1) Both the statements (A) and (B) are correct
(2) Only statement (A) is correct
(3) Both the statements (A) and (B) are incorrect
(4) Only statement (B) is correct

177. Consider the following features:
(a) Supported by incomplete cartilaginous rings.
(b) Lined by ciliated pseudostratified epithelium.
(c) Found within the lung lobe.
(d) Located below the secondary bronchi.
All of the above given features hold true for
(1) Tertiary bronchi (2) Alveolar ducts
(3) Terminal bronchiole (4) Alveoli

178. Choose the **correct** option w.r.t. a component of respiratory system which displays following features
(i) Double layered membrane formed by squamous epithelium.
(ii) Fluid present between two layers reduces friction on lung surface
(iii) Its outer layer remains in contact with thoracic wall
(iv) Its inner layer is in contact with lungs
(1) Visceral layer (2) Peritoneum cavity
(3) Visceral organs (4) Pleura

179. Trachea is a straight tube which divides into a right and left primary bronchi at the level of
(1) 2nd cervical vertebra (2) 5th thoracic vertebra
(3) 7th thoracic vertebra (4) 2nd lumbar vertebra

180. Which of the following helps in clinical assessment of pulmonary functions?
(1) ECG (2) Angiograph
(3) Sphygmomanometer (4) Spirometer

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